

Language Interface Layer (LIL): Qualitative Example

A) Raw Prompt (Baseline)

[implicit system boundary]

You are a helpful assistant. Answer concisely.

[implicit user boundary]

Explain representational stability in machine learning.

Extension (destabilizes tokenization)

Additional context that is not directly relevant to the question: The weather today is sunny with a high of 72 degrees. Stock markets closed higher on Tuesday amid e

- No explicit role markers
- Extension changes all token boundaries
- Repeated patterns not compacted

B) LIL Wire Format

Explicit structure with macro compaction:



Legend:

MAGIC/V1 = Header SYS/USR = Role marker M: name = Macro ref [] = Segment

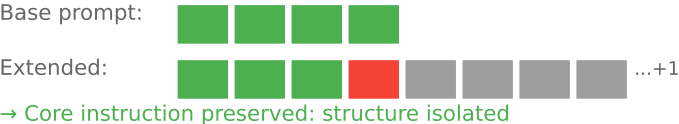
- ✓ Explicit role boundaries
- ✓ Macros compact 'helpful assistant' → 1 byte
- ✓ Extension isolated to USER segment
- ✓ Size: 101B → 97B (96%)

C) Tokenization Stability

Baseline (raw bytes):



LIL-stabilized:



Legend: ■ = stable ■ = changed ■ = new (extension)

Key insight: LIL explicit structure prevents extension
from destabilizing tokenization of core instruction.